Resolution, Smearing, and Unfolding in the Run-24 $p+p$ EMCal π^0/η Non-Linearity Study

A Reading and Teaching Report on B. Seidlitz's Working Directories

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Abstract

This report explains the three concepts that drive the sPHENIX Run-24 $p+p$ π^0/η EMCal calibration, non-linearity, and PPG12 isolated-photon work: **resolution**, **smearing**, and **unfolding**. Each concept is first taught from first principles, then tied line-by-line to the actual code that implements it in Blair Seidlitz's working areas (`/sphenix/u/bseidlitz/work/` and its GPFS mirror `/sphenix/user/bseidlitz/` — the two are the same content). Every file and directory inspected is named explicitly so the report can be retraced. A companion plain-text inspection log, `RESO_SMEAR_UNFOLD_LOG.txt`, lists the exact paths, sizes, and what was read.

Contents

1 Background and context of the study

The physics goal is a clean measurement of π^0 and η mesons (and isolated photons, PPG12) in $\sqrt{s} = 200$ GeV p+p collisions recorded by sPHENIX in Run 24. Both mesons are reconstructed through their two-photon decay channel,

$$\pi^0 \to \gamma\gamma \qquad (m_{\pi^0} = 0.134977 \text{ GeV}), \qquad \eta \to \gamma\gamma \qquad (m_\eta = 0.547862 \text{ GeV}),$$

by pairing electromagnetic clusters in the EMCal (CEMC) and forming the two-photon invariant mass

$$M_{\gamma\gamma} = \sqrt{2E_1E_2\,(1-\cos\theta_{12})}.$$

The measured peak position M_{fit} probes the *energy scale* and its *non-linearity* (does $M_{\text{fit}}/m_{\text{PDG}}$ stay flat versus p_T?), while the measured peak width $\sigma(M)$ probes the *energy resolution*.

To compare data with simulation fairly, the Geant4 MC must reproduce the detector's true performance. In practice the MC is slightly *too good* (its clusters are sharper than the real detector). The study therefore:

1. measures the resolution and energy scale in data via the meson peaks,
2. **smears** the MC so that simulated peak widths and positions match data,
3. propagates that tuned MC into efficiency and **unfolding** corrections for the final p_T spectra / cross-sections (PPG12).

The directories examined for this report:

- `/sphenix/u/bseidlitz/work/resolutionStudy/` — standalone toy-MC resolution/smearing demonstration.
- `/sphenix/u/bseidlitz/work/emcPerf/calo_emc_pi0_tbt/` — the production tower-by-tower π^0/η calibration module (`pi0EtaByEta.cc`), which contains the real smearing routine.
- `/sphenix/u/bseidlitz/work/emcPerf/macros/figMaker/` — `smear_compare.C` and `function_compare_wi` where the data/MC smearing functions are derived and stored.
- `/sphenix/u/bseidlitz/work/adjMCCalib/` — MC energy-scale fine-tuning CDB trees (`convSwitch.C`).
- `/sphenix/u/bseidlitz/work/ppg12/toymcunfold/macro/` — `resolutionsim.C` and `jet_pt_unfolding.C`, the RooUnfold examples.

2 Resolution — what it is and how it is defined here

2.1 Teaching: the meaning of resolution

Energy resolution is the fractional uncertainty with which the calorimeter measures a particle's energy. If a fixed-energy photon is sent into the EMCal many times, the reconstructed energies form a roughly Gaussian distribution of width σ_E about a mean E. The resolution is the ratio

$$\boxed{\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus c = \sqrt{\left(\frac{a}{\sqrt{E}}\right)^2 + c^2}}$$

where $\oplus$ denotes addition in quadrature. The two physically distinct terms are:

Stochastic term $a/\sqrt{E}$: from shower-sampling and photo-statistics. It *shrinks* as energy rises, so high-energy showers are measured proportionally better.

Constant term c: from calibration spread, light-collection non-uniformity, leakage, and dead material. It does *not* shrink with energy and therefore dominates at high E.

Because π^0/η widths come from *two* photons, the meson relative width $\sigma(M)/M$ inherits this same $a/\sqrt{E} \oplus c$ shape versus p_T and is the experimental handle used throughout these directories.

2.2 Where it is defined in code

The cleanest written-out definition is the standalone toy in /sphenix/u/bseidlitz/work/resolutionStudy/eta_decay_simulation_with_smearing.C. There the constant and stochastic terms are literal constants and are combined in quadrature exactly as above:

```
// energyResolution = constant term c = 12%
const double energyResolution = 0.12;
...
// stochastic term a/sqrt(E) with a = 15%
double energyDepSmear1 = 0.15 / TMath::Sqrt(photon1.E());
// quadrature sum c (+) a/sqrt(E) -> width of the Gaussian smear
double smearFactor1 = 1 + randGen.Gaus(0,
      TMath::Sqrt(energyResolution*energyResolution
                + energyDepSmear1*energyDepSmear1));
```

This is exactly the curve labelled in the plotting macro .../resolutionStudy/plot.C as

```
myText(0.65,0.30+0.5,1,"Toy MC 15/#sqrt{E} #oplus 12%");
```

i.e. $\sigma_E/E = 15\%/\sqrt{E} \oplus 12\%$. The companion file .../resolutionStudy/etaDecaySim.C is an earlier variant with $a = 20\%$, $c = 15\%$. The output `output.root` / `meson_graphs.root` feed `plot.C`, which fits each p_T slice with a Gaussian, forms σ^η/M^η, and overlays the toy curve on the p+p data points (`width_vs_pt.png`). **This is the operational definition of "resolution" in the study: the p_T dependence of the fitted meson relative width.**

3 Smearing — what it is and how it is done here

3.1 Teaching: the meaning of smearing

Smearing is the deliberate degradation of a simulated quantity so that the MC reproduces the (worse) resolution of real data. The truth-level MC photon energy E is replaced by

$$E \longrightarrow E \cdot f, \qquad f \sim \mathrm{Gaus}\big(1,\ \sigma_E/E\big),$$

a random multiplicative factor drawn from a Gaussian centred at 1 with width equal to the desired *extra* resolution. Smearing is the inverse mindset to resolution: resolution *measures* the blur; smearing *injects* the blur into an otherwise-too-perfect simulation. Position (angle) smearing works the same way but is *additive* on η and ϕ and centred at zero.

3.2 Toy-level smearing

In the toy .../resolutionStudy/eta_decay_simulation_with_smearing.C the photon 4-vector is scaled by the Gaussian factor while keeping its direction fixed (lines 69–70):

```
photon1.SetPxPyPzE(photon1.Px()*smearFactor1, photon1.Py()*smearFactor1,
                   photon1.Pz()*smearFactor1, photon1.E()*smearFactor1);
```

3.3 Production smearing: `pi0EtaByEta::smearPhot`

The real, analysis-grade smearing lives in /sphenix/u/bseidlitz/work/emcPerf/calo_emc_pi0_tbt/pi0EtaByEta.cc, function `smearPhot()` (lines 863–927). Key points:

- The energy- and position-resolution widths are *not* hard-coded constants; they are read as ROOT `TF1` functions of energy from `.../emcPerf/macros/figMaker/output/function_compare_wide.root` (opened in `Init()` at line 206, variable `kSmearParFile`).
- Energy smear: `Efac = rnd->Gaus(1.0, E_res)` with `E_res = f_energy_smear[idx]->Eval(E)`; clamped non-negative.
- Position smear: `Pfac_eta/phi = rnd->Gaus(0.0, P_res)` added to η, ϕ with ϕ wrapped to $(-\pi, \pi]$.
- Defaults if the file is missing: `E_res = 0.08` (8%), `P_res = 0.0001`.

```
float Efac = rnd->Gaus(1.0f, E_res); // energy smear factor
float Pfac_phi = rnd->Gaus(0.0f, P_res); // angular smear (phi)
float Pfac_eta = rnd->Gaus(0.0f, P_res); // angular smear (eta)
...
const float E_s = E * Efac;
sp.SetPtEtaPhiE(E_s/std::cosh(eta_s), eta_s, phi_s, E_s);
```

Six smearing variations (systematics). The integer `smear_idx` (0–5) selects a systematic variation: `base_smear_idx = idx%3` picks one of three energy/position function sets named in the code as `global_minimum`, `cE_0p08`, `cE_0p05` (i.e. different assumed constant terms $c = 0.08$, 0.05); and for `idx`$\geq$ 3 an *extra* 8% constant term is added in quadrature:

```
if (add_extra_const) E_res = std::sqrt(0.08f*0.08f + E_res*E_res);
```

These six histograms (`h_m_pt_smear1..6`, filled at lines 740–745) are how the smearing systematic is propagated.

3.4 Where the smear functions come from

The functions stored in `function_compare_wide.root` are produced by `/sphenix/u/bseidlitz/work/emcPerf/macros/figMaker/smear_compare.C`. This macro loads MC π^0/η mass-vs-p_T histograms (`h_m_pt_smear12/13/14`), fits each with the `MesonFitter` class, and compares the fitted mean ($M_{\text{fit}}/m_{\text{PDG}}$, the *non-linearity*) and relative width (σ/M_{fit}, the *resolution*) across smear variations labelled `A(-1%)/`$\sqrt{E}$, `A/`$\sqrt{E}$, `(A+1%)/`$\sqrt{E}$. The per-p_T spread (max$-$min)/average is taken as the smearing systematic. **Loop closed:** data widths $\Rightarrow$ choose smear function $\Rightarrow$ store `TF1` $\Rightarrow$ apply in `smearPhot` $\Rightarrow$ re-fit smeared MC $\Rightarrow$ verify it matches data.

3.5 Related: MC energy-scale tuning

Smearing fixes the *width*; a separate small *scale* correction fixes the *mean*. That lives in `/sphenix/u/bseidlitz/wor` `convSwitch.C` calls `scaleCDBTree(...)` to rescale the MC tower ADC→E calibration (`CEMC_MC_calib_default_v3` Scale tuning and width smearing together align MC to data in both peak position and peak width.

4 Unfolding — what it is and how it is done here

4.1 Teaching: the meaning of unfolding

A measured spectrum is the *true* spectrum convolved with detector effects (resolution/smearing, efficiency, bin migration). **Unfolding** is the statistical inversion that recovers the truth-level distribution from the measured one. Formally, with a *response matrix* $R_{ij} = P(\text{measured bin } i \,|\, \text{truth bin } j)$

built from MC,

$$\vec{m} = R\,\vec{t} \quad \Longrightarrow \quad \vec{t}_{\text{unfold}} = \mathcal{U}(R, \vec{m}),$$

where naive matrix inversion amplifies statistical noise, so iterative **Bayesian unfolding** (D'Agostini) is used instead — regularised by a small number of iterations. The MC that builds R must already be *smeared* to match data resolution; otherwise the response is wrong and the unfolded result is biased. This is the direct link from the smearing work to PPG12.

4.2 Toy unfolding and the resolution-mismatch bias

`/sphenix/u/bseidlitz/work/ppg12/toymcunfold/macro/resolutionsim.C` is a pure teaching toy. It loads `libRooUnfold.so` (from `/sphenix/user/egm2153/...`) and, for several pairs of (MC resolution, data resolution), builds a response from a smeared exponential p_T spectrum and unfolds:

```
RooUnfoldResponse *response = new RooUnfoldResponse(nbins,xlow,xup);
double recophotonpT = photonpT * r->Gaus(1, reso_mc); // build response (MC reso)
response->Fill(recophotonpT, photonpT);
...
double recophotonpT = photonpT * r->Gaus(1, reso_data); // "data" with a DIFFERENT reso
RooUnfoldBayes *unfold = new RooUnfoldBayes(response, hMeasPT, 2); // 2 iterations
TH1D *hRecoPT = (TH1D*)unfold->Hreco();
```

It then plots the **bias** (unfolded−truth)/truth. The pairs `reso = {(0.1,0.1),(0.1,0.11),(0.1,0.09),(0.05,0.` `(0.05,0.04)}` deliberately mismatch MC and data resolution to show how a poorly-tuned smearing inflates the unfolding bias — the quantitative argument for why smearing must be tuned first.

4.3 Realistic unfolding with Miss/Fake/Match

`/sphenix/u/bseidlitz/work/ppg12/toymcunfold/macro/jet_pt_unfolding.C` is the full machinery on real JetValidation trees. It demonstrates the bookkeeping that the toy omits:

- `response->Fill(reco,truth)` for matched ($\Delta R < 0.3$) pairs;
- `response->Miss(truth)` for truth jets with no reco match;
- `response->Fake(reco)` for reco jets with no truth match;
- a half-sample *closure test* (`resp_half`, `hMeasPTHalf`) to verify the procedure returns truth on independent MC;
- `RooUnfoldBayes(resp_full,hMeasPT,1)` with 1–4 iterations as the regularisation parameter.

The PPG12 `README.md` confirms this is the production strategy: "Bayesian unfolding (RooUnfold)", with energy-resolution *smearing variations* (`config_eres.yaml`) and unfolding iteration count listed explicitly among the systematic uncertainties.

5 How the three concepts fit together

1. **Resolution** is measured in data as $\sigma(M)/M$ vs p_T of the π^0/η peaks (`resolutionStudy/`, `figMaker/`).
2. **Smearing** injects exactly that much extra blur into MC so simulated peaks match data, via `pi0EtaByEta::smearPhot` using TF1s from `function_compare_wide.root`; six variations give the smearing systematic. Scale tuning (`adjMCCalib/`) aligns the mean.
3. **Unfolding** uses that smeared MC to build the response matrix and Bayesian-inverts the measured spectrum back to truth for the final cross-section (`ppg12/toymcunfold/`, PPG12 pipeline).

One-line glossary. *Resolution* = how blurry the detector is (measured). *Smearing* = adding that blur to MC (applied). *Unfolding* = removing the blur from a measured spectrum to recover truth (corrected).

All inspected files are read-only collaborator areas (owner `bseidlitz`). Exact paths, file sizes, and line references are recorded in `RESO_SMEAR_UNFOLD_LOG.txt` in this directory.